# CS 363M — Principles of Machine Learning
# Fall 2026

**Midterm exam**

Name: _____ EID: _____

Course modality (circle one): In-person Online

## Instructions

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*Read this page before you begin. Do not turn the page until you are told to start.*

- There are **5 questions** worth a total of **100 points**. Point values are shown beside each question and part.
- Budget your time according to the point values. Computational parts generally take longer than conceptual parts.
- **Each question begins on its own page. Write your answer for each part in the blank space directly below it, and label it** with its part number — 1.1, 1.2, and so on.
- **You must submit your cheat sheet with your exam.** Write your name on it and hand it in stapled behind the last page. An exam handed in without its cheat sheet will not be graded until the sheet is produced.
- One double-sided cheat sheet is permitted.
- **Show your work on every computational part.** An answer with no work receives no credit. Correct work with an arithmetic slip receives most of the credit, so write down the steps even when you are unsure of the final number.
- Round to four decimal places unless a part tells you otherwise. Leaving an answer as an unreduced fraction is fine.
- A basic or scientific calculator is permitted. Phones, laptops, tablets and smart watches must be off and out of sight for the whole period.
- If a part seems ambiguous, write down the assumption you are making and answer under that assumption. A clearly stated assumption will be graded fairly.
- Keep your eyes on your own paper. Suspected academic dishonesty is referred to Student Conduct and Academic Integrity.
- If you finish early you may hand in your exam and leave quietly. In the last ten minutes, please stay seated so you do not disturb the others.

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**Good luck.**

## 1. Evaluating classifiers (20 points)

A university wants to use an AI-text detector to check student essays. To test it, the detector is run on 400 essays whose authorship is known: 60 were written with AI and 340 by students. Of the 400 essays, the detector flags 50 as written with AI. Of these 50 flagged essays, 45 really were written with AI. Treat "written with AI" as the positive class.

**1.1 (4 points) Write down the confusion matrix.**

Answer 1.1

| | flagged | not flagged |
|---|---|---|
| written with AI | | |
| written by students | | |

**1.2 (2 points) What are precision and recall for this detector?**

Answer 1.2

**1.3 (2 points) Do we care more about improving precision or recall for this problem, and why?**

Answer 1.3

The detector also gives each essay a score between 0 and 1; a higher score means "more likely written with AI". Here are 8 essays, 4 written with AI and 4 by students:

| essay | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| score | 0.95 | 0.85 | 0.70 | 0.60 | 0.50 | 0.40 | 0.30 | 0.10 |
| written with AI? | yes | yes | no | yes | no | yes | no | no |

We say the detector flags an essay as AI-written if its score is over $t$.

**1.4 (10 points) The ROC curve starts at $(\mathrm{FPR}, \mathrm{TPR}) = (0, 0)$, when $t$ is above every score. Lower $t$ past one essay at a time. What are the next three points on the curve?**

**Answer 1.4**

**1.5 (2 points) Another detector has an AUC of 0.3 on the test essays. What does that tell you, and what simple change would improve it?**

**Answer 1.5**

## 2. Support vector machines (20 points)

**2.1 (10 points)** A soft-margin SVM returned $\boldsymbol{w} = (6, 8)$, $b = -20$.

| point | C | D | E |
|---|---|---|---|
| $x_1$ | 1.5 | 1.25 | 3 |
| $x_2$ | 1.5 | 1.5 | 0.5 |
| label | $+$ | $-$ | $-$ |

**(a) Compute $f(x)$ and the predicted class of each point.**

**(b) What is the width of the margin?**

**(c) Compute each point's distance to the decision boundary.**

**(d) Which points are on the margin? Which points are inside the margin? Which points are misclassified?**

**Answer 2.1**

**2.2 (4 points)** The soft-margin SVM solves

$$\min_{\boldsymbol{w},\, b,\, \boldsymbol{\xi}} \ \tfrac{1}{2}\|\boldsymbol{w}\|^2 + C\sum_i \xi_i \quad \text{subject to} \quad y_i(\boldsymbol{w}\cdot\boldsymbol{x}_i + b) \geq 1 - \xi_i, \ \ \xi_i \geq 0.$$

**(a) What does the slack $\xi_i$ measure, and why do we need it?**

**(b) Where does a training point lie, relative to the margin and the decision boundary, if $\xi_i > 1$?**

**Answer 2.2**

**2.3 (4 points)** Two soft-margin SVMs are trained on the same data, one with $C = 0.01$ and one with $C = 100$. For (a)–(c), circle one answer.

| | | |
|---|---|---|
| **(a) Which has the wider margin?** | $C = 0.01$ | $C = 100$ |
| **(b) Which has more support vectors?** | $C = 0.01$ | $C = 100$ |
| **(c) Which is more likely to overfit?** | $C = 0.01$ | $C = 100$ |

**(d) How would you choose $C$ in practice, and why?**

**Answer 2.3(d)**

**2.4 (2 points) You remove a training point that is not a support vector and retrain the SVM. What changes? Why?**

Answer 2.4

## 3. Nonlinear regression (20 points)

**3.1 (4 points)** kNN regression is trained on six points:

| $x$ | 1 | 2 | 4 | 5 | 7 | 8 |
|---|---|---|---|---|---|---|
| $y$ | 3 | 6 | 4 | 8 | 10 | 12 |

**Predict $y$ at $x = 4.4$ using (a) $k = 1$ and (b) $k = 3$.**

Answer 3.1

**3.2 (6 points) For each of the following, name a hyperparameter that controls how closely the model fits the training data. Which setting of it makes overfitting more likely?**

**(a) Regression tree**

Answer 3.2(a)

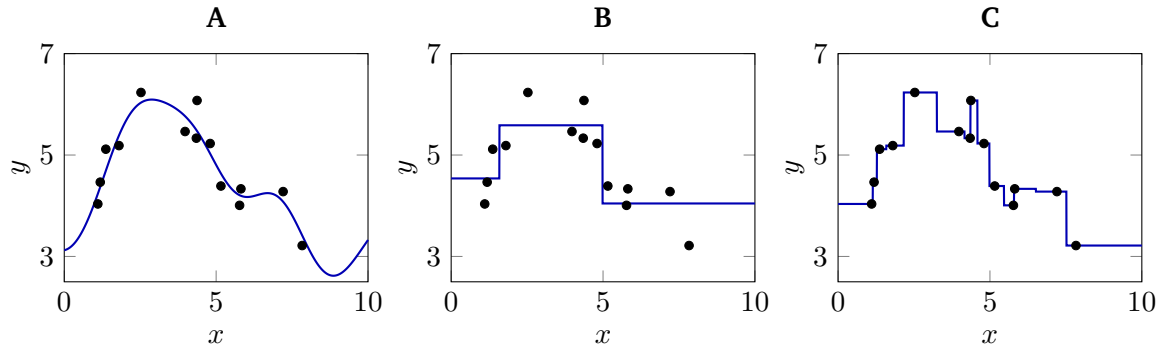
**(b) kNN regression**

Answer 3.2(b)

**(c) SVR**

Answer 3.2(c)

**3.3 (5 points) The figure shows three models fitted to the same data. Which panel shows a regression tree, which shows kNN with $k = 1$, and which shows SVR with a kernel? Explain each.**




Answer 3.3

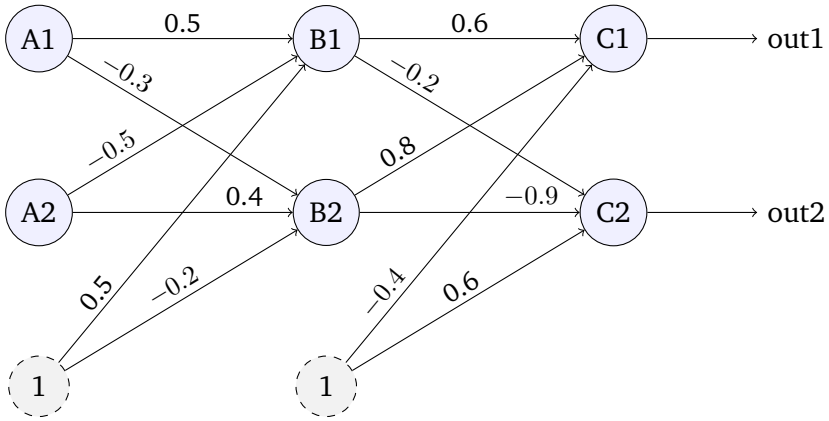
**3.4 (5 points) Suppose you use kNN to predict house prices from floor area (500 to 5,000 square feet) and number of bedrooms (1 to 6). Why might kNN do badly on these raw features? What would you do?**

Answer 3.4

## 4. Neural networks (30 points)

In this question, row $j$ of a weight matrix $\boldsymbol{W}^{(l)}$ holds the weights into node $j$ of layer $l$: $W_{ji}^{(l)}$ is the weight from node $i$ of the previous layer to node $j$. The bias of node $j$ is $b_j^{(l)}$, the weight on the input that is always 1.

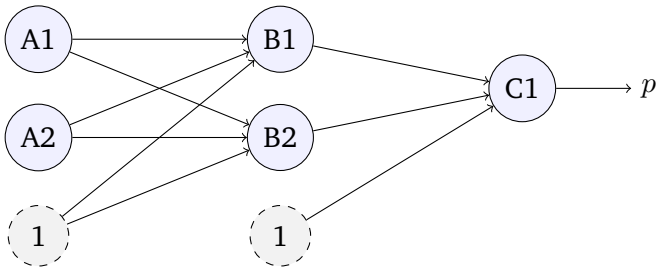
**4.1 (8 points)** In the network below, out1 represents Class A and out2 represents Class B. The hidden nodes B1 and B2 use the ReLU activation, $\mathrm{ReLU}(z) = \max(0, z)$. The output nodes C1 and C2 have no activation: each outputs its weighted sum. The network predicts the class whose output is larger.




**Show the forward pass, in matrix form, of the record with Feature1 $= 4$ and Feature2 $= 2$. How would this network classify the record (Class A or Class B)?**

**Answer 4.1**

**4.2 (14 points)** The network below has one hidden layer and one output. A record with inputs $A_1 = 2$, $A_2 = 1$ and label $Y = 1$ is passed in.




| | function | derivative |
|---|---|---|
| B layer: sigmoid | $f(z) = 1/(1+e^{-z})$ | $f(z)\,(1-f(z))$ |
| C layer: sigmoid | $p = C_1 = 1/(1+e^{-z_C})$ | (see the loss) |
| loss: cross-entropy | $E = -Y \log p - (1-Y)\log(1-p)$ | $\partial E/\partial z_C = p - Y$ |

Here $z_C$ is C1's input sum, and $p$ is the predicted probability that $Y = 1$. The weights during the forward propagation are

$$\boldsymbol{W}^{(1)} = \begin{pmatrix} 0.5 & -1.0 \\ 1.5 & 0.5 \end{pmatrix}, \quad \boldsymbol{b}^{(1)} = \begin{pmatrix} 0.5 \\ -1.0 \end{pmatrix}, \qquad \boldsymbol{W}^{(2)} = \begin{pmatrix} 1.0 & -2.0 \end{pmatrix}, \quad b^{(2)} = 0.5.$$

You do not need to forward propagate. The node output values during the forward propagation are

| A layer | B layer | C layer |
|---|---|---|
| $A_1 = 2$ | $B_1 = 0.62$ | $C_1 = p = 0.33$ |
| $A_2 = 1$ | $B_2 = 0.92$ | |

Use a learning rate of 1. Compute every derivative with the weights above, the ones used in the forward propagation.

**(a) Adjust $\boldsymbol{W_{11}^{(2)}}$, the weight from B1 to C1.**

**Answer 4.2(a)**

**(b) Adjust $W_{11}^{(1)}$, the weight from A1 to B1.**

**Answer 4.2(b)**

**4.3** Short answers (8 points in all).

**(a) (3 points) You train the same network twice with different learning rates and plot the training loss after each epoch. In run 1 the loss jumps up and down and ends higher than it started. In run 2 it goes down steadily, but very slowly. For each run, is the learning rate too large or too small? Why?**

Answer 4.3(a)

**(b) (2 points) A network classifies images into 3 classes. Why put a softmax on the three outputs instead of a separate sigmoid on each?**

Answer 4.3(b)

**(c) (3 points) A training set has 1,000 records. In one gradient-descent step, every weight in the network is updated once. How many steps does training take in 5 epochs (number of passes through the data) with (i) batch gradient descent, (ii) stochastic gradient descent (one record per step), and (iii) mini-batches of 50 records?**

Answer 4.3(c)

## 5. Ensemble methods (10 points)

**5.1 (5 points)** An SVM can only separate two classes. A clinic wants to classify chest X-rays into 4 classes: normal, pneumonia, COVID and tuberculosis.

**(a) With one-vs-rest partitions, how many SVMs do you train, and what labels does each one see?**

**(b) How is a new X-ray classified?**

**(c) With one-vs-one partitions, how many SVMs do you train?**

Answer 5.1

**5.2 (3 points) An ensemble of 25 classifiers is built, but each base classifier has error rate 0.6. What happens to the error of the majority vote, compared with a single classifier? Why?**

Answer 5.2

**5.3 (2 points) Random forests consider only a random subset of the features at each split. Why does this help the ensemble?**

Answer 5.3